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A Numerical Algorithm for Modeling Laminar Flows in an Annular Channel with Eccentricity

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Abstract—We propose a numerical algorithm for simulating steady laminar flows of an incompressible fluid in annular channels with eccentricity and rotation of the inner cylinder. This algorithm enables us to describe this class of flows for wide ranges of the annular channel and flow parameters. We present the implementation details and the results of testing this numerical method. For a series of flows in an annular clearance, we compare the numerical results to available analytic solutions and experimental data. In all cases under consideration the simulated data agree well with the available experimental, analytical, and numerical solutions.

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INTRODUCTION

The problem of a fluid flow in the channel between two cylinders is a classical hydrodynamics problem with wide practical applications. The examples of technical devices where these flows occur are heat exchangers, friction bearings, centrifuges, some types of mixers, drilling rigs, and others. The flow in the concentric channel is described by a well-known analytic solution [1]. In practice, however, this flow is usually complicated by the presence of eccentricity between the axes of outer and inner cylinders and the rotation of the latter. In the case of eccentric annular channels, no exact analytical solution is available, although many approximate and asymptotic solutions have been suggested.

Since a satifactory solution was absent until recently, the problem is still under active study. Some of the most valuable articles of the last decade on this topic are [2, 3] which present the results of experimental, analytical, and numerical studies. Firstly, experimental study was made of the influence of the angular velocity of the inner cylinder on the pressure drop. It is established that the pressure drop increases monotonely with the growth of the angular velocity of the inner cylinder, but the character of this dependence is nonlinear. Unfortunately, the authors restricted themselves to rather low Reynolds numbers Re_ω obtained from the angular velocity. The practical goals require a range of Reynolds numbers at least one or two orders higher.

Somewhat later, another systematic experimental and theoretical study of these flows appeared [4]. The axial velocity profiles in various regions are well obtained there for a series of regimes, and some data are presented on the pressure drop. Numerical results are obtained with rather high precision; nevertheless, they are still insufficient from the practical viewpoint. It was not possible to reconstruct the flow picture in various situations, find the laws governing the pressure drop, study the shear stress field, and so forth. Another important problem appearing here is related to the definition of the forces and momenta acting on the inner rotating cylinder. Therefore, despite the sixty-year-long history of studying data flows, the available material cannot fully represent all parameters of the flow as yet . For this reason,

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it is necessary to have tools by which we might be able to predict the characteristics of flows of this class with good precision and in a wide range of parameters.

The goal of this article is to create such tools. The algorithm to be developed must be suitable for modeling the laminar flows of Newtonian fluids in annular channels with eccentricity and rotation of the inner cylinder.

1. A MATHEMATICAL MODEL AND A NUMERICAL APPROACH

We consider isothermal laminar fully developed steady flow of an incompressible fluid with constant density. In this formulation, the flow is described by the Navier–Stokes equations

$$\nabla \cdot \rho \mathbf{v} = 0, \qquad \nabla \cdot \rho \mathbf{v}\mathbf{v} = -\nabla p + \nabla \cdot \mathbf{T}, \tag{1}$$

where ρ is the fluid density, $\mathbf{v}$ is the velocity vector, p is the pressure, $\mathbf{T} = \mu \mathbf{D}$ is the stress tensor, μ is the viscosity coefficient, and $\mathbf{D} \equiv D_{ij} = \left(\dfrac{\partial v_i}{\partial x_j} + \dfrac{\partial v_j}{\partial x_i} \right)$ is the deformation velocity tensor.

The simulated region is the annular channel formed by two smooth straight cylinders with circular cross-section. The radii of the inner and outer cylinders are equal to R_1 and R_2. The axis of the inner cylinder can be displaced by ΔR from the axis of the outer cylinder. In the sequel, we use the dimensionless eccentricity parameter $e = \Delta R/(R_2 - R_1)$. The inner cylinder may rotate about its axis with constant angular velocity ω. The no-slip condition holds on the surfaces of the cylinders.

In order to discretize the hydrodynamics equations we use the widely known control volume method. We obtain a finite difference analog of (1) by integrating over every control volume [5–8]. We use the structured meshes that are constructed in an orthogonal bipolar coordinate system using redistribution of nodes in the azimuthal direction. The application of the bipolar coordinate system for constructing the mesh enables us to perform an essentially homogeneous mesh refinement to the channel boundary. Since we consider a steady flow, the derivatives of the components of velocity along the z-axis (i.e., the channel axis) vanish. The degree of the mesh refinement in this direction does not substantially affect the precision of the mesh approximation; thus, in the direction of the flow it suffices to specify three mesh layers of the nodes forming two rows of control volumes. The length of the mesh in the direction of the z-axis is comparable to the outer diameter.

1.1. Discretization of the Transfer Equation

It is more convenient to construct the algorithm for solving a generalized linear convection-diffusion equation for the transfer of a scalar quantity ϕ, to which we can easily reduce the main conservation equations. The general form of the conservation equation for ϕ is

$$\nabla \cdot (\rho \mathbf{v} \varphi) = \nabla \cdot (\Gamma \nabla \varphi) + q_\varphi, \tag{2}$$

where Γ is the effective diffusion transfer coefficient, and q_ϕ is the source term. In order to obtain the balance relations of the finite volume method, integrate (2) over the volume Ω of a cell P and pass from the volume integral to an integral over the surface S:

$$\int\limits_S (\rho \mathbf{v} \varphi) \cdot d\mathbf{S} = \int\limits_S (\Gamma \nabla \varphi) \cdot d\mathbf{S} + \int\limits_\Omega q_\varphi \, dV.$$

The discretization of the volume and surface integrals rests on the mean value theorem. In result, as the average value over the volume we take the value in the cell center; while as the average value over a facet f, the value in the geometric center of f:

$$\sum_f (\rho \mathbf{v} \varphi)_f \cdot \mathbf{S}_f = \sum_f (\Gamma \nabla \varphi)_f \cdot \mathbf{S}_f + (q_\varphi)_P J, \tag{3}$$

where J is the cell volume and $\mathbf{S}_f$ the area vector of the facet. The summation here is over all facets of the control volume. The normal vector to the facet is directed away from the center of the control volume (see Fig. 1). To obtain the discretization of the source term we suppose a constant approximation on the

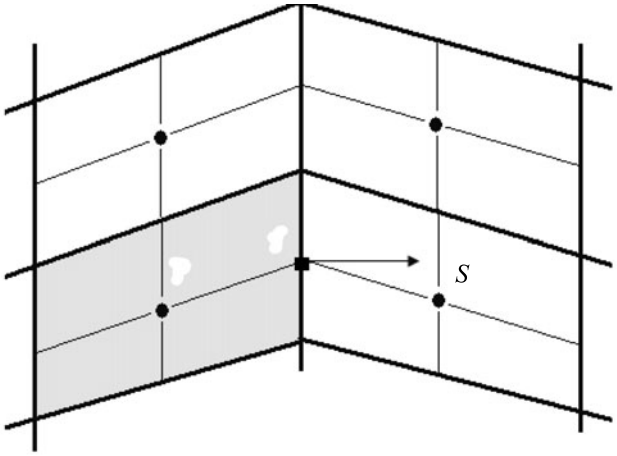


Fig. 1. Scheme of calculations of the fluxes through the facets of the control volume

control volume. In order to construct a conservative scheme and obtain algebraic equations on the mesh we have to approximate the values of the fluxes in the balance relation (3).

The first step in the discretization of fluxes is to calculate the gradients of the required quantity at the centers of the control volumes. We calculate the gradients, using the divergence theorem:

$$(\nabla\varphi)_P = \frac{1}{J}\sum_f \varphi_f \cdot \mathbf{S}_f.$$

We evaluate this quantity at the center of a facet by linearly interpolating the values at the centers of adjacent cells:

$$\varphi_f = \varphi_P(1-\lambda_f) + \varphi_S\lambda_f, \tag{4}$$

where the interpolation factor is defined as

$$\lambda_f = \frac{|\mathbf{r}_f - \mathbf{r}_P|}{|\mathbf{r}_f - \mathbf{r}_P| + |\mathbf{r}_f - \mathbf{r}_S|}.$$

Here $\mathbf{r}_P$, $\mathbf{r}_S$, and $\mathbf{r}_f$ are the radius vectors of the centers of the control volumes P and S and the geometric center of the facet f common to these control volumes (see Fig. 1).

1.2. Discretization of the Diffusion Term

The approximation of the diffusion term $\Gamma_f(\nabla\varphi)_f \cdot \mathbf{S}_f$ consists in interpolating the gradient from the values calculated in adjacent cells using (4),

$$\overline{(\nabla\varphi)_f} = (\nabla\varphi)_P(1-\lambda_f) + (\nabla\varphi)_S\lambda_f, \tag{5}$$

and isolating the derivative in the direction connecting the neighboring cells out of the normal derivative. This yields [5]

$$(\nabla\varphi)_f \cdot \mathbf{S}_f = A(\varphi_S - \varphi_P) + \overline{(\nabla\varphi)_f} \cdot (\mathbf{S}_f - A(\mathbf{r}_S - \mathbf{r}_P)), \tag{6}$$

where $A = \mathbf{S}_f^2/(\mathbf{S}_f \cdot (\mathbf{r}_f - \mathbf{r}_P))$.

We evaluate the diffusion transfer coefficient on a facet also by linear interpolation from the neighboring cells: $\Gamma_f = \Gamma_P(1-\lambda_f) + \Gamma_S\lambda_f$. The second term in (6) reflects the nonorthogonality of the mesh and leads to an extension of the difference stencil. The first term of the discretization of the diffusion term is accounted for implicitly, and we take the remaining part from the previous iteration and put it on the right-hand side of the system of equations under derivation.

On a facet of the control volume lying on the surface of the simulated region we apply a relation similar to (6) using the gradient at the center of the control volume:

$$(\nabla\varphi)_f \cdot \mathbf{S}_f = A(\varphi_f - \varphi_P) + (\nabla\varphi)_P \cdot \big(\mathbf{S}_f - A(\mathbf{r}_f - \mathbf{r}_P)\big), \qquad A = \mathbf{S}_f^2 / \big(\mathbf{S}_f \cdot (\mathbf{r}_f - \mathbf{r}_P)\big).$$

We determine the value of φ_f at the center of the facet from the boundary conditions.

1.3. Discretization of the Convective Term

The problem of finding the convective part of the total flux reduces to evaluating the required function on the facets of the control volume given its values at the nodes,

$$(\rho\mathbf{v}\varphi)_f \cdot \mathbf{S}_f = (\rho\mathbf{v} \cdot \mathbf{S})_f \varphi_f = \dot{m}_f \varphi_f,$$

where $\dot{m}_f$ is the mass flux through the facet. We evaluate the velocity vector at the center of the facet using some linear interpolation of type (4):

$$\mathbf{v}_f = \mathbf{v}_P(1 - \lambda_f) + \mathbf{v}_S \lambda_f.$$

The choice of an interpolation method between the nodes of the mesh determines the properties of the resulting difference scheme. We can express the scheme for approximating the convective term in a general form as

$$\begin{gathered} \dot{m}_f \varphi_f = \dot{m}_f^+ \varphi_f^+ + \dot{m}_f^- \varphi_f^-, \\ \dot{m}_f^+ = \max(\dot{m}_f, 0), \qquad \varphi_f^+ = \varphi_P + \frac{1}{2}\Psi\big(r_f^+\big)(\varphi_S - \varphi_P), \\ \dot{m}_f^- = \min(\dot{m}_f, 0), \qquad \varphi_f^- = \varphi_P + \frac{1}{2}\Psi\big(r_f^-\big)(\varphi_S - \varphi_P). \end{gathered} \tag{7}$$

The arguments of the restriction function $\Psi(r)$ are

$$r_f^+ = \frac{2(\nabla\phi)_P(\mathbf{r}_S - \mathbf{r}_P)}{(\phi_S - \phi_P)} - 1, \qquad r_f^- = \frac{2(\nabla\phi)_S(\mathbf{r}_S - \mathbf{r}_P)}{(\phi_S - \phi_P)} - 1.$$

The restriction function $\Psi(r)$ is constructed so that, on the one hand, we obtain a high order of approximation and on the another, the validity of the boundedness criterion for the difference scheme is guaranteed. In simulation, we used the Leonard's countercurrent scheme QUICK [9] of the second approximation order: $\Psi(r) = (3 + r)/4$. From the expression (7) for the flux we select the term that corresponds to the countercurrent scheme of the first order of approximation:

$$(\dot{m}_f \varphi_f)^{k+1} = \big(\dot{m}_f^+ \varphi_P + \dot{m}_f^- \varphi_S\big)^{k+1} + \big(\dot{m}_f \varphi_f - \big(\dot{m}_f^+ \varphi_P + \dot{m}_f^- \varphi_S\big)\big)^k. \tag{8}$$

In the iterative process, the first term in parentheses in (8) is accounted for implicitly. The second term in (8) raises the order of the scheme by extending the computational stencil.

In order to ensure the diagonal domination of the coefficient matrix in the difference equation (3), for every control volume we add the term (if the problem converges then it tends to zero):

$$\big(\varphi_P^{k+1} - \varphi_P^k\big) \sum_f \dot{m}_f.$$

In result, the expressions for the convective flux at the nodes of P and S become as follows:

$$\begin{aligned} (\dot{m}_f \varphi_f)^{k+1} - \dot{m}_f \varphi_P^{k+1} &= \dot{m}_f^- (\varphi_S - \varphi_P)^{k+1} + \big[\dot{m}_f \varphi_f - \big(\dot{m}_f^+ \varphi_P + \dot{m}_f^- \varphi_S\big)\big]^k, \\ (\dot{m}_f \varphi_f)^{k+1} - \dot{m}_f \varphi_S^{k+1} &= \dot{m}_f^+ (\varphi_P - \varphi_S)^{k+1} + \big[\dot{m}_f \varphi_f - \big(\dot{m}_f^+ \varphi_P + \dot{m}_f^- \varphi_S\big)\big]^k. \end{aligned}$$

Summing all fluxes through the facets of the control volume (3), we obtain a difference representation of the original differential equation. Collecting the coefficients of the variables at the centers of the control volumes, we find the algebraic relation among the values of the required function at the center of a control volume and neighboring control volumes:

$$\sum_f a_f(\varphi_S - \varphi_P) + Q_P = 0.$$

The first term involves the values of the neighboring cells, and in the iterative algorithm it is treated implicitly. The second term Q_P involves the diffusion flux term (6), related to the nonorthogonality of the mesh, and the terms raising the approximation order of the convective flux (8). We calculate Q_P from the values obtained at the previous iteration.

The above discretization of the fluxes and source term yields a system of linear algebraic equations with the coefficient matrix a_f, diagonal coefficients a_P, and right-hand side Q_P:

$$\sum_f a_f \varphi_S + Q_P - \varphi_P a_P = 0, \qquad a_P = \sum_f a_f. \tag{9}$$

The matrix enjoys the diagonal domination which is a necessary condition for convergence of the iterative method for solving linear systems.

In order to improve the convergence, we introduced the lower relaxation procedure [6], which enables us to raise the stability of the numerical solution on account of decreasing the rate of convergence. For the transfer equation the relaxation procedure is introduced directly into (9) via a relaxation parameter α, $0 < \alpha < 1$, by the following method of strengthening the diagonal:

$$\frac{1}{\alpha} a_P \varphi_P^{k+1} = \sum_f a_f \varphi_S^{k+1} + Q_P + \frac{1-\alpha}{\alpha} a_P \varphi_P^k.$$

It is obvious that the strengthening of diagonal domination in the matrix improves convergence. Moreover, without this relaxation procedure we are unable to obtain a converging solution. It is convenient to express the lower relaxation procedure in an algebraically equivalent form basing on a nonstationary statement with the introduction of a local pseudotime parameter τ:

$$\tau = \frac{\rho J}{a_P} \frac{\alpha}{1-\alpha}, \qquad \left(a_P + \frac{\rho J}{\tau} \right) \varphi_P^{k+1} = \sum_f a_f \varphi_S^{k+1} + Q_P + \frac{\rho J}{\tau} \varphi_P^k. \tag{10}$$

Therefore, the parameter τ depends on the geometry of the control volume P and the local values of velocity and characterizes the time required for convective and diffusive changes in the required quantity in this control volume. Our simulations demonstrated that the coefficient of relaxation on the components of velocity equal to 0.8 suffices to obtain the converging results while preserving a high rate of convergence.

1.4. SIMPLE-Similar Procedure for Splitting the Velocity and Pressure Fields

In the incompressible fluid approximation, any direct relation between the fields of pressure and velocity is absent, which leads to additional computational difficulties. Several approaches are available for overcoming this: the methods related to the introduction of artificial compressibility [14]; the methods based on using the flow and vorticity functions [1, 13]; as well as the methods based on applying the Poisson equation to the pressure or pressure correction [5, 6]. In our algorithm, the relation between the velocity and pressure fields is realized using a SIMPLE-similar procedure on superimposed meshes [6, 7].

The iterative calculation of the velocity and pressure fields consists of three stages at every iteration. First, basing on the solution to the linearized second equation in (1), we calculate an intermediate value of the velocity $\mathbf{v}^*$:

$$\nabla \cdot (\rho \mathbf{v}^k \mathbf{v}^*) - \nabla \cdot (\tau^*) = -\nabla p^k,$$

where $\mathbf{v}^k$ and p^k are the values of velocity and pressure at the previous iteration. The linear algebraic system obtained by discretizing the linearized equation is solved iteratively by the incomplete factorization method D-ILU [10] in which only the diagonal terms are factorized. The intermediate velocity field found at the first stage in general fails to satisfy the continuity equation. In order to satisfy the continuity condition we make a correction step at which we solve an equation on the pressure correction p' and determine pressure at the next iteration:

$$\nabla \cdot (\tau \nabla p') = \nabla \cdot (\rho \mathbf{v}^*), \qquad p^{k+1} = p^k + p'.$$

Here τ is a coefficient possessing the dimension of time, and for the SIMPLE-C algorithm it is defined [7] in (10). The resulting difference analog of the equation for pressure correction is solved using the algebraic multigrid method [11]. In the multigrid algorithm, we make presmoothing and postsmoothing using the incomplete factorization method D-ILU. Finally, at the last stage we make the correction to the velocity determined as

$$\mathbf{v}^{k+1} = \mathbf{v}^* - (\tau/\rho)\nabla p'.$$

With this correction, the velocity field satisfies the continuity equation and is divergence-free in the case of a constant density. We choose zero values at the zeroth iteration: $\mathbf{v}^0 = 0$ and $p^0 = 0$.

In order to avoid any pressure field oscillations due to superimposed variables, we follow the approach of Rhie and Chow in which a "monotonizer" is introduced [12] on the right-hand side of the pressure correction equation. On each facet, to the value of mass flux we add some correction that is proportional to the difference between the pressure gradient on the facet and the value interpolated from the values of the gradients at the centers of adjacent nodes (see (5)):

$$\dot{m}_f = \rho \mathbf{v}_f \cdot \mathbf{S}_f - \mathrm{RCh}\, \rho \tau_f A\big[(p_S - p_P) - \overline{(\nabla p)_f} \cdot (\mathbf{r}_S - \mathbf{r}_P)\big].$$

Here RCh is a coefficient allowing us to control the share of the Rhie–Chow correction, which we set to 0.1 [7].

1.5. Procedure of Flow Rate Correction

The flow in question is usually determined by a constant mass flow rate Q of the fluid. Moreover, some pressure gradient forms along the channel, ensuring this flow rate. In order to obtain steady flow in a channel, we use an iterative procedure for correcting the flow rate. The procedure consists of the three stages: At the first stage, we determine the velocity on the boundary using the values obtained at the previous iteration in the boundary control volumes $\mathbf{u}^k$, starting from the soft boundary condition

$$\frac{\partial}{\partial \mathbf{n}} \cdot \mathbf{v} = 0, \qquad \mathbf{v}_f^* = \mathbf{v}_P^k;$$

i.e., we extrapolate the boundary values of the velocity components from the inner region. Here $\mathbf{n}$ is the normal to the boundary of the control volume in question, $\mathbf{v}_P^k$ is the velocity at the center of the boundary control volume P, and $\mathbf{v}_f^*$ is the velocity on the boundary of P.

At the second stage, we correct the velocity thus-calculated on the boundary by accounting for the given mass flow rate on the boundary:

$$\mathbf{v}_i^k = \mathbf{v}_i^* + \frac{\mathbf{n}_i\Big(Q - \sum\limits_f \rho_f \mathbf{v}_f^* \cdot \mathbf{S}_f\Big)}{\rho_i \sum\limits_f \mathbf{n}_f \cdot \mathbf{S}_f}.$$

Here the subscripts enumerate the facets, the summation is over all facets belonging to the region on which the flow rate Q is defined (incoming or outgoing), and $\mathbf{S}_f$ is the vector whose length equals the area of the facet and whose direction coincides with that of the normal vector $\mathbf{n}$. At the last stage, the adjusted velocities are used as the Dirichlet boundary conditions for the equation of motion.

We can use this approach to obtain the steady flows in the channels. The procedure of correcting the fluxes enables us to construct a fully developed flow on a short fragment of the channel without having to simulate long channels with a zone of flow stabilization. This approach is close to the method of periodic boundary conditions.

We determine pressure on the boundaries by interpolation:

$$p_f = p_P + (\nabla p)_P \cdot (\mathbf{r}_f - \mathbf{r}_P).$$

On the solid walls, the velocity satisfies the usual no-slip and impermeability conditions, while the pressure satisfies the condition of vanishing of the normal derivative: $\dfrac{\partial p}{\partial n} = 0$.

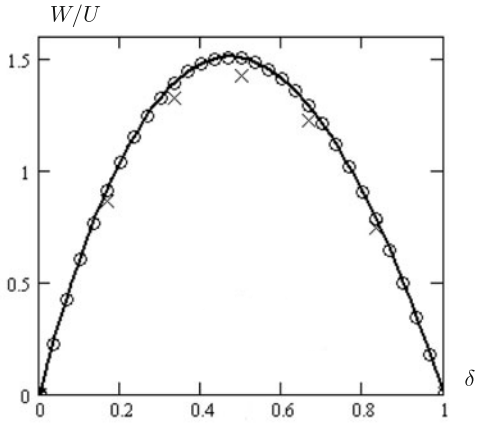


Fig. 2. Velocity profile in the concentric annular channel: solid lines show the exact solution; × shows simulation on 7×40 mesh; and ○ shows simulation on 30×80 mesh

2. TESTING THE ALGORITHM

Basing on the algorithm in Section 1, we created a software package and tested it for two years by solving many problems of this class either allowing analytic solutions or coming with available experimental data. Also we made comparisons to numerical solutions [4]. In this process of testing, we varied the flow parameters and the annular channel geometry in wide ranges. We present here only some of the data obtained. In all cases the iterative process was stable and yielded converging results. The criterion for stopping the iterations was the relative iterative change in the main fields settling at the level of less than 10^{-6}. The result obtained by iterations is independent of the initial distribution, the lower relaxation coefficient, and the choice of a method for solving the linear algebraic system.

2.1. Flow in Pipes

We started with testing the algorithm on the analytical solutions in the cases of flows in pipes with circular, square, and triangular cross-sections (for instance, see [13]). We used the meshes of various spatial resolution. In all cases, we established good agreement of the simulated data with the analytic values. The decrease of mesh size substantially improved the results. However, the precision of about 0.5% is already achieved on rather coarse meshes. Here and below we indicate the relative error of computations. One of the important characteristics in practice is the pressure drop along the pipe. In a circular pipe, the precision of 0.1% is achieved on a 20×20 mesh, while in the pipes with square and triangular cross-sections, on the meshes 50×50 and 100×100 respectively.

2.2. Flow in a Concentric Channel

An exact analytical solution is available for this problem as well [1]. In test simulations, we varied the Reynolds number Re and the ratio Θ of the radii of the inner and outer cylinders of the annular channel. In all cases, we managed to achieve high precision and convergence rate on rather coarse meshes. Fig. 2 shows the comparison of the analytical and simulated velocity profiles obtained on two different meshes for the flow at $\mathrm{Re} = 800$ and $\Theta = 0.5$. The velocity profiles are related to the average velocity of the flow determined from the flow rate. The horizontal axis indicates the dimensionless distance between channel walls (we pass to dimensionless variables relative to the width of the annular clearance.) The continuous curve corresponds to the analytic solution, the circles to a 30×80 mesh, and the crosses to a mesh with 7×40 nodes in the radial and tangential direction respectively. The numerical solution obtained on a mesh with 30×80 nodes coincides with the analytical solution with the precision of 0.5%. We determined the pressure difference with the same precision.

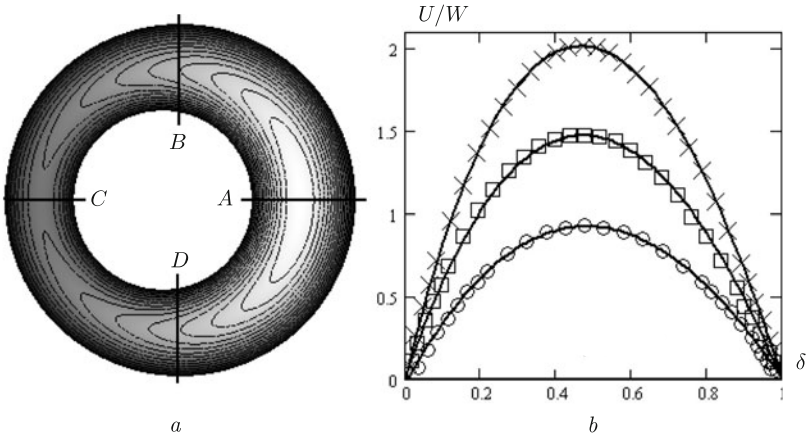


Fig. 3. Isoclines (a) and profiles of the axial components of velocity (b) in various regions of the channel for $\Theta = 0.506$, $\mathrm{Re} = 105$, and $e = 0.2$: $\times$ for region A; $\square$ for region B; and $\circ$ for region C

2.3. Flow in an Annular Channel with Eccentricity

This problem admits no exact analytical solution, but its practical importance is so great that it was studied experimentally many times. The latest and most precise experiments were reported in [4]. The velocity profiles were measured for an annular channel with the ratio of radii of inner and outer cylinders equal to $\Theta = 0.506$ for three values of eccentricity: $e = 0.2$, 0.5, and 0.8. Below in Fig. 3 and in this section, we present a comparison of our simulated data with the experimental data of [4].

Fig. 3, b shows the comparison of the simulated (solid lines) and experimental (symbols) velocity profiles for $e = 0.2$ and three different regions of the annular channel, whose locations are shown in Fig. 3, a. The axial Reynolds number was equal to 105. The inner cylinder was not rotating. The numerical solution is obtained on the mesh with 40×120 nodes and agrees well with the experimental data.

A numerical solution was also constructed in [4] by a version of the finite volume method on a nonorthogonal mesh. Calculations were made using a second-order central-difference scheme. Our results agree with the data of [4] up to the construction.

Fig. 4 illustrates a similar comparison of the simulated and experimental data for $e = 0.8$ (the remaining parameters are the same as in the simulation presented in Fig. 3, n).

Substantial eccentricity leads to the distribution of velocity in the channel becoming strongly inhomogeneous. The maxima of the axial velocity in the narrowest and the widest parts of the channel differ by a factor of more than 20. The maximal deviation of the simulated data from the experimental data was at most 1% with the exception of those points where we can see that the experimental data clearly involves with quite large errors. This experimental data is obtained in region C (see Fig. 4) near the walls, where measurement is difficult. It is worth noting that in this region the calculated data of [4] also substantially differ from the experimental data but agree with ours up to construction.

2.4. Flow in an Annular Channel with Eccentricity and Rotation of the Inner Cylinder

Finally, consider the most complicated case of a flow in a channel with eccentricity and rotation of the inner cylinder. Consider rotation with angular velocity 13 rad/s, corresponding to the Taylor number $\mathrm{Ta} = 3000$, where $\mathrm{Ta} = (\rho\omega/\mu)^2(R_1 - R_2)^3$. Furthermore, the axial Reynolds number is equal to 115, and $e = 0.5$. Fig. 5 shows a comparison of the simulated and experimental profiles of the axial and tangential velocities in various regions of the channel. The simulated velocity profiles are depicted as solid curves, and experimental profiles as symbols. We ran the simulation on a mesh with 40×120 nodes. As above, we observe a good agreement of the results of simulation and experiment; they coincide up to the precision of the experimental data.

Fig. 6 illustrates comparison of the simulated pressure difference with the experimental data of [2]. Here we present the dependence of the dimensionless pressure difference in an annular channel ($\Theta = 0.5$,

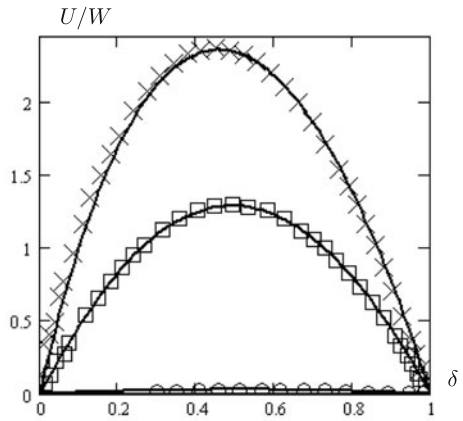


Fig. 4. Profiles of the axial components of velocity in various regions of the channel for $\Theta = 0.5$, $\mathrm{Re} = 105$, and $e = 0.8$: $\times$ for region A; $\square$ for region B; and $\circ$ for region C

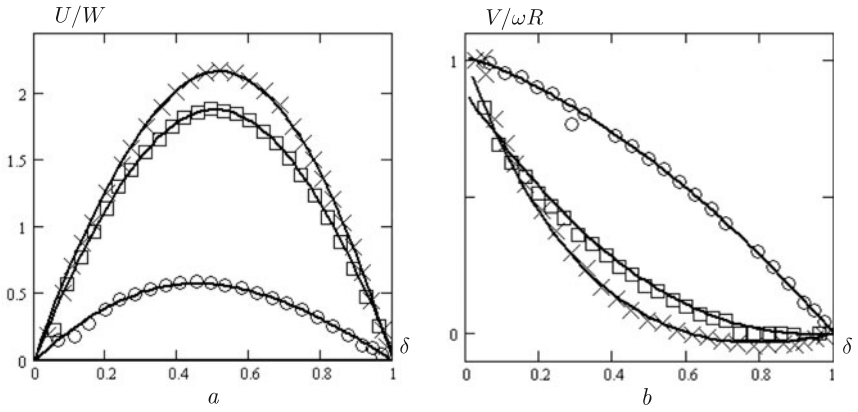


Fig. 5. Profiles of the axial (a) and tangential (b) velocities in various regions of the channel for $\Theta = 0.5$, $\mathrm{Re} = 115$, $e = 0.5$, and $\mathrm{Ta} = 3000$: $\times$ for region A; $\square$ for region B; and $\circ$ for region C

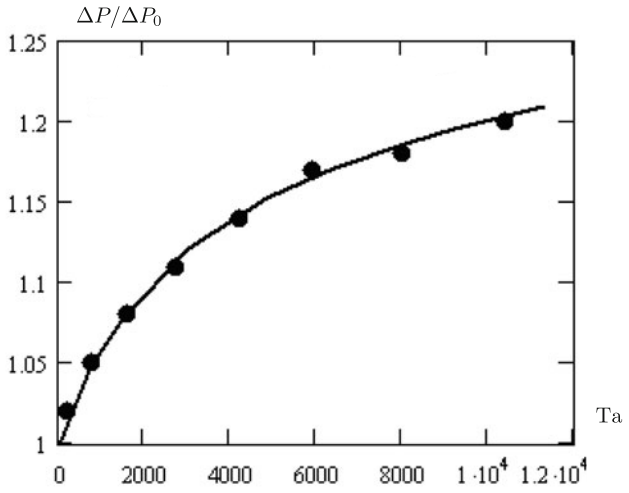


Fig. 6. Dependence of the pressure difference in an annular channel on the Taylor number: $\bullet$ for experiment [3] and solid lines for simulation

$\mathrm{Re} = 115$, and $e = 0.5$) on the Taylor number. We pass to dimensionless variables relative to the pressure difference in the channel with a fixed inner cylinder ΔP_0. Again, the results of simulation agree with the experimental data up to the precision of the latter.

CONCLUSION

Summarizing our tests, we can conclude as follows: The results of this article about modeling by the numerical algorithm for simulating the laminar flows of the Newtonian fluids in annular channels with eccentricity and rotation of the inner cylinder agree well with the available analytical solutions and existing experimental and numerical data. Therefore, the proposed algorithm enables us to model the flow for a wide range of flow parameters and channel geometries. In fact, the algorithm enables us to model the flow for an arbitrary geometry of the annular channel. In our simulations, the ratio of radii varied in the range $0.139 \le \Theta \le 0.91$, while the eccentricities were essentially arbitrary ($0 \le e < 1$). The flow parameters (fluid flow rate, its density, viscosity, angular velocity of the inner cylinder) also varied in quite wide ranges. However, we should keep in mind that this flow is unstable with respect to the external even small perturbations. This means that transition to turbulence happens in some range of the flow parameters, and we should require higher precision of the algorithm. The flow is particularly critical to the angular velocity of the inner cylinder. It is known that transition to turbulence in a cylindrical Quette flow (in the absence of enforced flow of the fluid) happens at rather low Reynolds numbers.

To apply this algorithm for solving practically important problems would require generalizing it to the case of non-Newtonian fluid flows. In practice we have to deal with the so-called power-law fluids, Bingham fluids, Herschel–Bulkley fluids, Robertson–Stiff fluids, and others. The principal complexity in modeling the viscoelastic flows is related to the existence of singularity in those regions where the rate of deformation vanishes. For viscoelastic fluids with nonzero limiting stress τ_0 in the flow region $|\mathbf{T}| < \tau_0$ (the solid body region) the rate of shear tends to zero, and for this reason the effective viscosity of the fluid increases unboundedly.

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